

Session 2



Advanced HTML & Forms

Tables · Forms & Inputs · Semantic HTML · Audio/Video · iframe

HTML Tables

HTML Forms

Semantic Tags

Multimedia

iframe

Prepared & Presented by

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Technical Trainer | Frontend Development Bootcamp 2026

About the Presenter

Utkarsh Kushwaha

Full Stack Developer | Technical Speaker | Community Mentor

Experienced in building responsive and interactive web applications using modern technologies and actively involved in technical communities, workshops, and student learning initiatives.

TECHNICAL SKILLS



HTML5



CSS3



JavaScript



React.js



Node.js

express Express.js



MongoDB



Git & GitHub



REST APIs



Responsive Web Design

Connect me on
LinkedIn



Connect me on
GitHub



Learning Objectives

By the End of This Session, Students Will Be Able To:



Create and structure HTML tables



Design webpages using semantic HTML tags



Build interactive HTML forms



Embed audio and video in webpages



Use different input elements effectively



Integrate external content using iframe



Understand labels and form controls



Develop user-friendly and structured webpages

Session Outcome: Students will gain practical understanding of advanced HTML elements used in real-world webpages.

Session Agenda



HTML Tables

Rows, columns, merge cells



Forms & Input Elements

text, email, radio, checkbox



Labels & Form Controls

Accessibility & UX



Semantic HTML Tags

header, nav, section, footer



Audio & Video

<audio> and <video> tags



iframe Embedding

YouTube, Maps, external pages

Learning Outcome: Develop structured and user-friendly web pages using advanced HTML elements.

Quick Revision

Previously Learned in Session 1

- ➔ Basic HTML Structure
- ➔ Headings (<h1>—<h6>)
- ➔ Paragraphs <p>
- ➔ Ordered & Unordered Lists
- ➔ Links <a href>
- ➔ Images

CODE

```
<h1>Welcome to HTML</h1>
<p>This is a paragraph.</p>

<ul>
  <li>Item One</li>
  <li>Item Two</li>
</ul>

<a href="page.html">Click Here</a>

```



HTML Tables

Organizing Data into Rows & Columns

Introduction to HTML Tables

What is a Table? HTML tables organize data into rows and columns.

`<table>`

The container for the entire table

`<tr>`

Table Row — each horizontal row

`<th>`

Table Header — bold & centered by default

`<td>`

Table Data — regular cell content

Table Structure Preview

Name	Age	Grade
Rahul	20	A
Priya	21	B+

Basic Table Example

CODE

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
    <th>City</th>
  </tr>
  <tr>
    <td>Rahul</td>
    <td>20</td>
    <td>Jabalpur</td>
  </tr>
  <tr>
    <td>Priya</td>
    <td>21</td>
    <td>Indore</td>
  </tr>
</table>
```

OUTPUT

Name	Age	City
Rahul	20	Jabalpur
Priya	21	Indore

<tr> = Defines each Table Row

<th> = Header cell (bold by default)

<td> = Data cell content

border = Adds visible borders

Advanced Table Features

colspan

`colspan="2"`

Merge cells horizontally

rowspan

`rowspan="2"`

Merge cells vertically

cellpadding

`cellpadding="5"`

Space inside each cell

cellspacing

`cellspacing="3"`

Space between cells

CODE

```
<td colspan="2">Total</td>
<td rowspan="2">Score</td>
```



HTML Forms

Collecting User Input Efficiently

Introduction to HTML Forms

Forms allow websites to collect input from users — login, signup, search, and more.

CODE

```
<form action="submit.php" method="POST">
  <!-- form elements go here -->
</form>
```

Common Input Types:

text

email

password

radio

checkbox

number

submit

reset

Enter your login credentials

Username:

Password:

Basic Form Structure

CODE

```
<form>

  <label>Name:</label>
  <input type="text" placeholder="Enter name">

  <br><br>

  <label>Email:</label>
  <input type="email" placeholder="Enter email">

  <br><br>

  <input type="submit" value="Submit">

</form>
```

`<form>`

Container for all form elements

`<label>`

Text describing the input field

`<input type="text">`

Single-line text field

`<input type="email">`

Email field with format validation

`<input type="submit">`

Button to submit the form data

Input Types in HTML

Input Type	Purpose	Example Use
<code>text</code>	Normal single-line text	Username, Name fields
<code>email</code>	Email with format check	user@example.com
<code>password</code>	Characters are hidden	Login passwords
<code>number</code>	Only numeric values	Age, Phone number
<code>checkbox</code>	Select multiple options	Interests, Features
<code>radio</code>	Select only one option	Gender, Plan type
<code>file</code>	Upload a file	Profile picture, CV
<code>submit</code>	Sends form data	Submit button

Text, Email & Password Inputs

CODE

```
<!-- Text Input -->
<input type="text"
  placeholder="Enter your name">

<!-- Email Input -->
<input type="email"
  placeholder="user@example.com">

<!-- Password Input -->
<input type="password"
  placeholder="Enter password">
```

text

Accepts any text. No built-in validation.

email

Validates format user@domain.com on submit.

password

Characters shown as •••• for security.

Radio Buttons & Checkboxes

CODE

```
<!-- Radio Buttons (Select ONE) -->
<input type="radio" name="gender" value="male"> Male
<input type="radio" name="gender" value="female"> Female
<input type="radio" name="gender" value="other"> Other

<!-- Checkboxes (Select MULTIPLE) -->
<input type="checkbox" name="lang" value="java"> Java
<input type="checkbox" name="lang" value="python">
Python
<input type="checkbox" name="lang" value="js">
JavaScript
```

🔴 Radio Button

Use when user must choose **ONLY ONE** option from a group.
All radio buttons in a group must share the same name attribute.

☑️ Checkbox

Use when user can select **MULTIPLE** options simultaneously.
Each checkbox works independently.

Dropdown & Textarea

CODE

```
<!-- Dropdown Menu -->
<label>Select Country:</label>
<select name="country">
  <option value="in">India</option>
  <option value="us">USA</option>
  <option value="uk">UK</option>
</select>

<!-- Multi-line Text Area -->
<label>About You:</label>
<textarea rows="4" cols="40"
  placeholder="Write something...">
</textarea>
```

<select>

Creates a dropdown menu. Wrap individual choices inside <option> tags.

<option>

Each option in the dropdown. Use value attribute to submit the right data.

<textarea>

Multi-line text input for longer content like comments, bio, or messages.

Labels & Form Controls



Better Accessibility

Screen readers read labels aloud for visually impaired users



Easy Clicking

Clicking the label text also activates the input field



User-Friendly Forms

Users instantly know which label belongs to which field

CODE

```
<!-- Without Label (BAD practice) -->
<input type="text">

<!-- With Label using 'for' and 'id' -->
<label for="username">Username:</label>
<input type="text" id="username">

<!-- The 'for' attribute must match 'id' -->
<label for="email">Email:</label>
<input type="email" id="email">
```

Label → Input Connection

```
<label for="name">Name</label>
```

↓ *for matches id*

```
<input id="name" type="text">
```

Mini Project: Student Registration Form

CODE

```
<form>
  <label for="name">Name:</label>
  <input type="text" id="name">
  <br><br>

  <label for="email">Email:</label>
  <input type="email" id="email">
  <br><br>

  <label for="course">Course:</label>
  <select id="course">
    <option>BCA</option>
    <option>B.Tech</option>
    <option>MCA</option>
  </select>
  <br><br>

  <label>Gender:</label>
  <input type="radio" name="g"> Male
  <input type="radio" name="g"> Female
  <br><br>

  <label>Interests:</label>
  <input type="checkbox"> Web Dev
  <input type="checkbox"> AI/ML
  <br><br>

  <input type="submit" value="Register">
</form>
```

Form Elements Used:

- ✓ text input
- ✓ email input
- ✓ select dropdown
- ✓ radio buttons
- ✓ checkboxes
- ✓ submit button



Semantic HTML

Building Meaningful Webpage Structure

Semantic vs Non-Semantic HTML

Semantic tags clearly describe the meaning and purpose of the content inside them.

Semantic Tags

`<header>`

`<nav>`

`<main>`

`<section>`

`<article>`

`<aside>`

`<footer>`

Non-Semantic Tags

`<div>`

`<span>`

`<div class='nav'>`

`<div class='head'>`

`<div id='main'>`

`<b>`

`<i>`

Semantic Tags in Practice

CODE

```
<header>
  <h1>My Portfolio</h1>
</header>

<nav>
  <a href="#home">Home</a>
  <a href="#about">About</a>
</nav>

<main>
  <section>
    <h2>About Me</h2>
    <p>I am a developer.</p>
  </section>

  <article>
    <h2>My Blog Post</h2>
    <p>Article content here.</p>
  </article>
</main>

<footer>
  <p>&copy; 2026 Utkarsh</p>
</footer>
```



Better SEO

Search engines understand your page structure better



Better Readability

Other developers can read and understand code easily



Accessibility

Screen readers navigate semantic pages correctly



Clean Structure

Code is organized and maintainable long-term



Audio & Video

Embedding Multimedia in HTML

Audio & Video in HTML

CODE

```
<!-- Audio Tag -->
<audio controls>
  <source src="music.mp3" type="audio/mpeg">
  Your browser does not support audio.
</audio>
```

CODE

```
<!-- Video Tag -->
<video controls width="400" height="250">
  <source src="movie.mp4" type="video/mp4">
  Your browser does not support video.
</video>
```

Key Attributes:

controls

Shows play/pause/volume UI

autoplay

Starts playing automatically

loop

Repeats the media endlessly

muted

Starts muted (needed for autoplay)

width/height

Sets display size for video

<source>

Specifies media file path & type

Embedding with <iframe>

CODE

```
<!-- Basic iframe -->
<iframe src="https://www.example.com"
  width="600" height="400">
</iframe>

<!-- Embed YouTube Video -->
<iframe
  src="https://www.youtube.com/embed/VIDEO_ID"
  width="560" height="315"
  allowfullscreen>
</iframe>

<!-- Embed Google Maps -->
<iframe
  src="https://maps.google.com/maps?q=Jabalpur"
  width="600" height="450">
</iframe>
```

Real-World Uses:



YouTube Videos



Google Maps



External Websites



Ads & Widgets

Best Practices in HTML



Use semantic tags

`<header>, <nav>, <section>, <footer>`



Write clean & indented code

Indent nested elements by 2 spaces



Always use labels in forms

Connect with `for` & `id` attributes



Use meaningful attribute names

`id='studentName'` not `id='s1'`



Add alt text to images

`<img alt="Profile Photo">`



Test forms in the browser

Check that submission & validation works

Today We Learned



HTML Tables

Rows, Columns, Merge Cells



HTML Forms

Form Container & Structure



Input Types

text, email, radio, checkbox...



Semantic HTML

Meaningful Tags & Better SEO



Audio & Video

<audio> & <video> Embedding



iframe

Embed External Content

Key Takeaway: Advanced HTML helps create interactive, structured, and user-friendly webpages.



Thank You!

Questions? I'm happy to help!

Practice Task:

*"Create a complete Student Registration Form using Semantic HTML.
Include a table for marks, a form with all input types, and embed a YouTube video using iframe."*

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